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# "We define X as …": A local grammar of definition in linguistics research articles and its pedagogical value

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ABSTRACT

Local grammars have been shown to be useful for accounting for discourse acts (e.g., exemplification, evaluation) in academic writing. Focusing on the specific discourse act of defining, this study develops a local grammar of definition in Linguistics research articles and further discusses its implications and applications for EAP writing research and pedagogy. Using a corpus compiled from published research articles, the study searched for a set of lexicogrammatical markers to retrieve instances of definition and the subsequent analyses identified 36 local grammar patterns of definition. The pedagogical implications and applications are then discussed in relation to EAP course design and the development of local grammar informed EAP teaching materials. The study concludes that systematically developing local grammars of discourse acts of importance in academic writing is a worthwhile undertaking and the findings of such research can usefully inform EAP writing pedagogy.

## 1. Introduction

Multiple studies have observed that there are many discourse acts, defined here as an equivalent of rhetorical or communicative functions such as exemplifying and defining, that are performed more frequently in academic texts than in other genres (e.g., Paquot, 2008, 2010). Such discourse acts play a crucial role in the construction and dissemination of specialist knowledge across academic subjects and also contribute to the persuasiveness and accessibility of academic texts (Hyland, 2007; Kääntä et al., 2018; Morton, 2020). In other words, the appropriate performance of these discourse acts is an essential skill for successful academic writing in all disciplines. However, English L2 or novice writers were often found to have difficulty in performing such discourse acts appropriately. This has been primarily attributed to the limited repertoire of the lexicogrammatical resources that they have at their disposal (e.g., Hinkel, 2004; Hyland & Tse, 2004; Paquot, 2008). Thus, to help English L2 or novice writers to overcome this obstacle, there is a pressing need to systematically document and describe the linguistic resources associated with a given discourse act.

To meet this rising demand, recent studies have shown that local grammars are particularly useful for unpacking and documenting the realisations of pragmatic or discourse acts (e.g., Barnbrook, 2002; Cheng & Ching, 2018; Su, 2017; Su & Wei, 2018; Su & Zhang, 2020). Briefly, local grammar refers to an approach to language and discourse studies that "seeks to account for, not the whole of a language, but one meaning only'' (Hunston, 2002a, p. 178) or, to put it more simply, a local grammar "is always a grammar of a discourse function" (Hunston & Su, 2019, p. 597; see Section 2). Following previous studies, the present study attempts to develop a

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local grammar of definition in Linguistics research articles and to further discuss its pedagogical implications and applications. It is part of a larger project which aims to systematically develop local grammars of discourse acts of importance in academic writing and to adequately explore the potential applications of these local grammars in English for Academic Purposes (EAP) writing pedagogy. It will be further shown in the present study that local grammars are useful for accounting for discourse acts in academic texts and that findings of local grammars of discourse acts can fruitfully inform EAP writing pedagogy.

The remainder of this paper is organised into five sections. Section 2 briefly discusses local grammars and reviews previous studies on the discourse act of defining. Section 3 introduces the corpus and the data retrieval procedures employed in the current investigation. Section 4 presents the detailed local grammar analyses of definition, followed by Section 5 which discusses the pedagogical implications and applications of local grammars of discourse acts in EAP writing pedagogy. Section 6 concludes this paper.

## 2. Local grammars of discourse acts in academic writing

### 2.1. Local grammar

As noted in Section 1, each local grammar seeks to account for one discourse function or act. Previous studies have developed local grammars of, for instance, evaluation (Hunston & Sinclair, 2000; Hunston & Su, 2019), definition (Barnbrook, 2002), disclaiming (Cheng & Ching, 2018), and exemplification (Su & Zhang, 2020), among many others. These studies show that one defining feature of local grammar is its use of context-specific functional terminologies, rather than the traditional grammatical terms (e.g., Subject, Predicate), to analyse the corresponding syntactic elements. This distinguishes it from general grammars (see also Hunston & Su, 2019; Su & Wei, 2018). For example,[1]

(1) I define political correctness as the elevation of sensitivity over truth.

In Table 1 we present the general grammar analysis and in Table 2 the alternative local grammar analysis.

The general grammar analysis, as presented in Table 1, shows clearly the grammatical relationship between the syntactic elements. However, it says nothing about the pragmatic function of the sentence as definition. This is consistent with Barnbrook and Sinclair's (2001, p. 243) argument that "[t]he general grammar would be seen as a structurally oriented grammar, capable of performing labelled bracketing on the sentences but not able to work out their functions" and that "general grammars, including those that call themselves 'functional', do not as yet explain how a definition defines" (ibid, p. 273). In contrast, the alternative local grammar approach analyses each element in terms of its pragmatic meaning and thus reveals clearly the pragmatic function of the sentence as definition. It is in this respect that local grammars have been considered particularly useful for investigating pragmatic or discourse acts (Su & Wei, 2018), as will be further demonstrated in the present study.

The rationale for employing a local grammar approach to account for discourse acts in written academic texts can be further justified as follows. First, early studies have argued that, compared with general grammars, the local grammar descriptions of language use have several advantages. For instance, Hunston and Sinclair (2000, p. 101) point out that local grammar analysis "would be more simple, more precise, and more useful than an analysis using a general grammar". Hunston (2002b, p. 157) further adds that "once the necessary patterns have been identified, the elements of the local grammar are more useful than elements of a general grammar are". Barnbrook and Sinclair (2001, p. 273), based on their study of defining language in dictionaries, make an even stronger argument that "[e]xperiment will tell us whether the definition grammar is always superior to the general grammar, […] *The likelihood is that such a specialised grammar will outperform a general grammar*" (emphasis added). Likewise, Butler (2004, p. 158) asserts that "rather than a single general grammar, we might end up with a set of local grammars for particular areas defined by their communicative functions in the discourse". The point here is that local grammars are a useful alternative approach to describe language in use, especially for the description of discourse or pragmatic acts (see also Barnbrook, 2002; Bednarek, 2008; Cheng & Ching, 2018; Hunston & Su, 2019; Su, 2017; Su & Zhang, 2020).

More importantly, studies have noted that the resulting local grammar descriptions could facilitate English language teaching (cf. Su, 2020). For instance, Hunston and Su (2019) have discussed the applications of local grammars in designing English teaching materials and argued that local grammars could contribute to developing a 'thesaurus-like constructicon' for English language teaching. In the EAP context, Su and Zhang (2020) proposed a sample coursebook unit to demonstrate how local grammars of discourse acts (in their case 'a local grammar of exemplification') can be applied to develop EAP teaching materials; and Zhang and Su (2021) further presented a local grammar based teaching procedure to show how local grammars can be applied in EAP pedagogical practice.

Aligning with previous studies, the present study attempts to develop a local grammar of definition by drawing on data from a corpus compiled from Linguistics research articles (see Section 4 for a discussion about the corpus). Its aims are twofold. On the one hand, it aims to document and describe the lexicogrammatical resources and the associated local grammar patterns that are available to perform the specific discourse act of defining, thereby contributing to the overarching goal of developing systematically local grammars of discourse acts of importance in academic writing. On the other hand, it further explores the pedagogical implications and applications of local grammars of discourse acts in the teaching of academic writing, thereby attempting to offer a possible alternative approach to EAP writing pedagogy.

[1] All the examples unless otherwise noted were taken from the corpus used in the present study. For the sake of clarity, however, some of the examples may have been simplified.

**Table 1**
General grammar analysis.

| Subject | Predicate | Object | Complement |
|---|---|---|---|
| *I* | *define* | *political correctness* | *as the elevation of sensitivity over truth.* |

**Table 2**
Local grammar analysis.

| Source-internal[a] | Hinge | Definiendum | Definiens |
|---|---|---|---|
| *I* | *define* | *political correctness* | *as the elevation of sensitivity over truth.* |

[a] The terminologies used for the local grammar analysis of definition are discussed in detail in Section 4.

### 2.2. Definition in academic discourse

The present study focuses on the specific discourse act of defining which, according to Dalton-Puffer (2013, p. 234), has the specific communicative intent of telling others "about the extension of an object of specialist knowledge". In Hyland's model of code glossing, definitions are considered as cases of explanations, i.e., "situated clarifications which elaborate the meaning of a preceding unit to make a concept more accessible by providing a gloss or a definition" (Hyland, 2007, p. 274). In a similar vein, Triki (2019, p. 105) states that "[d]efinitions are instances of elaboration used by writers in order to explain words, ideas, concepts, etc.". Drawing on insights from these aforementioned studies, we tentatively define definition as a discourse act that is performed to clarify abstract notions or concepts and/or to elaborate technical terms, as shown in examples (2)–(4).

(2) A refusal is a speech act in which a speaker denies to engage in an action proposed by the interlocutor.
(3) Lexicalized words were defined as those that have a conventional Chinese translation equivalent or correspond to a fixed item in Chinese.
(4) Task structuring comprised verbalizations of how participants were focusing their attention or otherwise planning their approach to lesson tasks.

We focus on definition in research articles because definitions are considered as representation of disciplinary knowledge (Clarence & McKenna, 2017) and, more importantly, consensus has been reached that definition is of particular significance in scientific discourse. For example, Trimble (1985) points out that definition is one rhetorical function that is typically performed in English science and technology discourse. Wilkins (1986, p. 53) similarly argues that definitions are "obviously a feature of scientific and other academic forms of writing". Ädel (2006) has also found that defining is one of the common discourse functions in argumentative essays and, in a similar vein, Swales and Feak (2012, p. 67) observe that "[d]efinitions are common in student papers and published papers". More recently, Triki (2019, p. 115) underlines that "definitions constitute a core function in academic writing", and Nashaat-Sobhy and Llinares (2020, p. 11) further note that "defining is a common requirement in the curriculum in all disciplines". In a nutshell, definition is pervasive in various forms of academic discourse across disciplines.

Explanations have also been offered to account for the crucial roles definition plays in scientific discourse. For instance, Swales (1971, p. 66) suggests that "[d]efinitions occur frequently in many types of scientific writing because it is often necessary to define certain operations, substances, objects or machines" (cited in Pearson, 1998, p. 89). Halliday and Martin (1993, p. 230) argue that "[d]efinitions are important because they relate known common-sense terms or previously defined technical terms to new technical ones". Moreover, Dalton-Puffer (2007) points out that offering definitions could also help subject matter learning. Specifically, she argues that "thinking explicitly about definitions would deepen students' understanding and enrich the semantic representation of central concepts of the subjects they are studying, thus making an important contribution to subject matter learning" (Dalton-Puffer, 2007, p. 138). To sum up, discourse acts such as defining and exemplifying are ubiquitous because they "are central in the construction of everyday knowledge and knowledge associated with specialised domains, among them academic disciplines" (Kääntä et al., 2018, p. 694).

Given its importance, efforts have been made to explore definitions, especially in dictionaries and educational settings. For example, Cumming et al. (1994) evaluated the effectiveness of four approaches to word definition in dictionaries and found that learners prefer sentence format of definition with usage examples. Scott and Nagy (1997) examined upper elementary students' ability to understand different formats of definitions provided in dictionaries for unfamiliar verbs, exploring how such an ability related to students' learning of vocabulary. Barnbrook (2002) presented probably by far the most comprehensive study which developed a local grammar of definition by focusing on definition sentences in *Collins Cobuild Student's Dictionary* (Sinclair, 1990). He also demonstrated that the parser developed on the basis of a local grammar of definition could be effectively applied to automatically analyse definition sentences of words. Nevertheless, since Barnbrook's (2002) local grammar of definition focused on definitions in dictionaries, our effort to develop a local grammar of definition in written academic discourse could complement Barnbrook's pioneering work, thereby contributing to a more adequate description of the basic language function of definition and its linguistic realisations.

In educational settings, Flowerdew (1992) explored definitions in lectures of biology and chemistry given by native speaker lecturers to non-native speaker students. His investigation shows that definition occurs rather frequently in such lectures, approximately

one definition per 1 min 55 s on average and that there is a great discrepancy between definitions offered by lecturers and those presented in EAP textbooks. From the perspective of systemic functional grammar (Halliday, 1994), Harvey (1999) analysed definitions in English technical discourse and explored the degree of equivalence constructed in theses definitions. Temmerman (2009) compared teacher definitions, pupil definitions and jointly formulated definitions and found that teacher definitions are crucial in helping students grasp the meanings and usages of words. Adopting a multimodal conversation analysis approach, Kääntä et al. (2018) explored how a physics teacher accomplished definition and definition-related actions in lectures. They observed that different resources were coordinated to illustrate key concepts, which helped the students to understand the physical law being instructed. In another study, Nashaat-Sobhy and Llinares (2020) analysed and compared CLIL students' written definitions of historical terms in Spanish (L1) and English (L2) of two different historical fields across two educational levels. The results showed that while there were no differences in the frequency and type of definitions across languages, the field and the educational level of the students appeared to have influenced frequency, types, and composition of their definitions in English.

The brief review presented above suggests that, although definitions are common in academic writing, many previous studies have either generally looked into definition in scientific discourse or primarily focused on definitions in dictionaries and educational settings. However, to our knowledge, very few studies have accounted for the discourse act of defining in prestigious academic genres such as research articles. One notable exception is Triki (2019) who, drawing on extracts from a corpus of Linguistics and Computer Science research articles, examined the metadiscursive aspect of definitions and found that writers of the two disciplines prefer different strategies for definition (e.g., Linguistics writers use definitions from external sources more frequently than their counterparts of Computer Science). To gain a fuller understanding of this discourse act in high-stakes academic genres, more studies from different theoretical or methodological perspectives are certainly needed. Moreover, we also observe that the majority of previous studies are essentially qualitative and have provided little, if any, quantitative information of the distinct realisational patterns of definition. We argue, however, that the provision of quantitative information of the realisations of a given discourse act is valuable in that such information could usefully inform EFL learners of the typicality of each distinct realisation of that discourse act. This could in turn facilitate the teaching of discourse acts in EAP writing courses; for example, different realisational strategies of a given discourse act could be taught according to the degree of typicality (see Section 5 for further discussion).

In addition, amid the current 'publish or perish' culture and in the English as a lingua franca context, academics, especially English L2 and novice writers, face great pressure in writing English research articles for publication. One of the major obstacles they have, as reported in previous studies, relates to the performance of discourse acts. For instance, in their study of expressing acknowledgements in MA and PhD dissertations, Hyland and Tse (2004, p. 260) argue that "little is known of their structure and expression, and students are often left to their own devices when writing them". Regrettably, this appears to apply to learners' performance of other discourse acts as well. In the case of exemplification, for example, Hinkel (2004, p. 297) notes that although giving examples is a powerful, prevalent thesis support strategy in constructing persuasive essays in English, college-level L2 writers rarely employ this strategy successfully in compliance with the guidelines in L2 composition instruction. Similarly, Paquot (2008) observes that English L2 writers' reservoir of exemplifying phraseologies are far more limited than their English L1 counterparts.

To recapitulate, while definition is of paramount importance in academic writing and has attracted much attention, our review suggests that there is still ample room for further investigation into definition, especially in high-stakes academic genres such as research articles. Meanwhile, English L2 and novice writers in general have difficulty performing appropriately specific discourse acts in academic writing, definition being no exception. It is therefore worthwhile and necessary to thoroughly explore the lexicogrammatical resources that are available to perform these discourse acts in academic contexts. To that end, our suggested solution is to systematically develop local grammars of discourse acts of importance in academic writing (Su & Zhang, 2020), and in this study we zoom in on a local grammar of definition. We hope to further show that each local grammar can not only adequately describe the target discourse act, but also offer an inventory of the lexicogrammatical resources and the associated discourse-semantic patternings that English L2 and/or novice writers can avail of to perform that specific discourse act, thereby contributing to EAP writing research and pedagogy.

## 3. Corpus and data retrieval

For the current investigation, we compiled a corpus by collecting research articles published between 2010 and 2019 in the journal *Applied Linguistics* (https://academic.oup.com/applij/). We selected research articles as the academic genre of focus because they are probably the most important type of academic texts that can represent "expert performances" (Bazerman, 1994); and our decision to sample research articles published in *Applied Linguistics* was driven by its relatively long history (it was established in the 1980s) and its high impact – in August 2021 it was ranked first in Linguistics with an impact factor of 5.741. These properties allow us to claim that the texts we sampled are highly representative of academic writing in the field of Linguistic and can thus be considered as valid points of references for the teaching of academic writing in this discipline. An overview of the corpus, which we have called the Corpus of

**Table 3**
Meta-information of CORAL.

| No. of texts | No. of types[a] | No. of tokens[b] | Average length (by tokens) |
|---|---|---|---|
| 314 | 2,792,711 | 3,598,204 | 11,459.25 |

[a] In corpus linguistics, 'type' refers to the total number of different words in texts.
[b] 'token' refers to the total number of words in texts.

Research Articles in Linguistics (hereafter CORAL), is given in Table 3.

Two additional issues concerning the data need to be addressed. The first relates to our focus on the discipline of Linguistics. Despite the criticism that EAP studies have overly focused on this discipline (Swales, 2019), our decision to examine Linguistics research articles relates to our overall project design. In the larger project that this study belongs to, we intend to explore not only how writers of published research articles, but also how advanced Chinese English L2 student writers perform discourse acts in high-stakes academic genres such as degree dissertations (see also Su et al., 2022), so as to offer insights into EAP teaching in EFL contexts. This overall project design does not give us any other choice in terms of discipline, as programmes in other disciplines (except for a few EMI programmes) do not require students to write their degree dissertations in English in mainland China. The other issue has to do with our decision to neglect the author status in the sampling process. That is, we did not differentiate articles authored by English L1 writers from those by EFL writers. This is because all articles published in the target journal have not only been rigorously peer reviewed, but also carefully edited after being accepted, and as such the language used in these articles is acceptable to the disciplinary community.

We then uploaded the corpus to Sketch Engine (Kilgarriff et al., 2004) for data retrieval. As our investigation focuses on what Darian (2003, p. 54) calls 'overt definition' which "contains a formal marker (lexical, typographic) identifying it as a definition", we used a pre-determined set of lexical items and syntactic constructions to search for instances of definition in our corpus. This set of items is built upon previous studies which have shown that definitions are typically signalled by lexical or structural markers. For example, Flowerdew (1992, p. 215) noted that "[m]ost definitions have some sort of syntactic and/or lexical signalling, the copula being the most important syntactically, and *call/called* and *means* the most important lexically". Pearson (1998) discussed comprehensively the lexical (e.g., *mean, define, refer*) and syntactic signals (e.g., *a(n) X is Y …, I/we define X as …*) of explicit definitions, and Ädel (2006) further examined definitional verbs such as *define*, *mean*, *denote*, etc. used for definitions. In a similar vein, Triki (2019) used a set of definitional markers as the starting point to explore how definitions were used in research articles of Linguistics and Computer Science.

Building on these previous studies, we identified as comprehensive as possible a set of items and used them to search for and retrieve instances of definition in CORAL. For the sake of clarity, these items were tentatively classified into four broad categories: definitional construction, definitional nouns, definitional verbs, and definitional linking adverbials. Items in each category and their quantitative information are presented in Table 4.

With regard to retrieving data by searching for these items, a few points need to be clarified. First, searching for these items may return hits which are not instances of definition. A good case in point is the search of the 'X BE Y' construction. In Sketch Engine, the script for searching for this construction is "[tag = "N.*"][lemma = "be"][ ]{0,2}[tag = "N.*"]", allowing up to two words that may occur between the copula 'BE' and 'Y'. The search, however, returns many hits which do not have the illocutionary force of definition (e.g., *Chang is a native speaker …*). Other similar examples are *that is* and *i.e.* We found that, while both *that is* and *i.e.* can be used to define, many instances co-occurring with them are paraphrasing, i.e., restating (e.g., *would age of onset determine L2 attainment, that is, in particular, would late age of onset of SLA preclude …*), or exemplifying (e.g., *Emoticons (i.e.:) …*), rather than defining. Such instances were excluded at the later stage of local grammar analysis. Secondly, we observed that some instances of definition may contain more than one item included in the pre-determined list (e.g., *The term 'context' has been used to refer to many different aspects …*). Such instances were analysed and tallied only once, although they could be retrieved by searching for either item. Lastly, it should be acknowledged that our method of data retrieval has an obvious limitation, that is, searching for these pre-determined items can by no means exhaust all the instances of definition in a corpus; for example, it cannot capture covert definitions, i.e., those which were not

**Table 4**
Markers of definition and their frequency in CORAL.

| | Markers | Raw frequency | Normalised frequency[a] |
|---|---|---|---|
| Definitional construction | 'X BE Y' construction | 10,273 | 2855.04 |
| Definitional nouns | term (as noun) | 810 | 225.11 |
| | concept | 868 | 241.3 |
| | definition | 661 | 183.7 |
| | notion | 562 | 156.19 |
| Definitional verbs | consider | 1642 | 456.34 |
| | refer | 1228 | 341.28 |
| | define | 776 | 215.66 |
| | consist of | 438 | 121.73 |
| | mean (as verb) | 434 | 120.62 |
| | seen as | 301 | 83.65 |
| | comprise | 259 | 71.98 |
| | call (as verb) | 184 | 51.14 |
| | known as | 82 | 22.79 |
| | regard (as verb) | 35 | 9.73 |
| | term (as verb) | 5 | 1.39 |
| Definitional linking adverbials | i.e. | 1096 | 304.6 |
| | that is | 495 | 137.57 |
| | BE related to | 205 | 56.97 |
| | BE concerned with | 152 | 42.24 |

[a] 'Normalised frequency' refers to the frequency that was normalised per million words. This applies throughout the paper.

signalled by any lexical markers (e.g., *This is particularly so in commercial ELT, a lucrative global service industry* …). Nevertheless, as the list has covered as fully as possible those items that have been recognised by previous studies as signalling the discourse act of definition, we are confident that searching for these items would allow us to identify a substantial set of instances of definition in our corpus.

## 4. Towards a local grammar of definition

As discussed in Section 2.1, local grammars use context-specific and transparent functional terminologies to analyse corresponding syntactic elements. The identification of a set of such terminologies for the subsequent local grammar analysis of definition is thus crucial. In fact, a few early studies have also attempted to analyse definition by using functional terms. For example, Flowerdew (1992) suggested 'Term', 'Genus', and 'Differentia' in his discussion of definition in academic lectures; Barnbrook (2002) used 'Definiendum', 'Hinge', and 'Definiens' to explore dictionary definitions; and Darian (2003) adopted 'Term', 'Class word', 'Relator', and 'Limiting features' to analyse definition. In a more recent study, Triki (2019) employed such terminologies as 'Defined unit', 'Definition marker' and 'Defining unit' to investigate definitions in research articles of Linguistics and Computer Science. Drawing on insights from the aforementioned studies, we performed a preliminary analysis of the data we retrieved, based on which eight terminologies were identified for the subsequent local grammar analysis of definition. The eight terminologies are listed and succinctly explained in Table 5.

Two of these terminologies might need further explanation. One is the term 'Definiens'. In some cases, the 'Definiens' is realised by a nominal phrase indicating a superordinate category which, in Flowerdew's (1992) term, is labelled 'Genus'. This type of 'Definiens' is normally followed by a clause that specifies its limiting features, which we label as 'Supporting statement'. We did so because, while we agree that 'Genus' is a useful label for such nominal phrases, we had the impression that this term may sometimes cause confusion and cannot suitably account for 'Definiens' without 'Supporting statement'; for instance, it would be more appropriate to label *to a raw number of occurrence* as 'Definiens' than as 'Genus' in *Frequencies refer to a raw number of occurrences*. As such we use 'Definiens' to label the elements which clarify how a term or concept is being defined, be it a Genus or a nominal phrase that by itself adequately illustrates the term being defined. Structurally, in our analysis, 'Supporting statement' is normally realised by a postmodifier of the 'Definiens', either in the form of a finite or non-finite clause. The other term is 'Hinge'. In some cases, there may be more than one element that realises 'Hinge'. For instance, in *They defined a LRE as 'any part of a dialogue where the students talk about* …, both *defined* and *as* should be, strictly speaking, considered as 'Hinge'; however, for the purpose of not to overcomplicate the analysis, we labelled *defined* as 'Hinge' while considered the whole sequence *as any part of a dialogue* as realising 'Definiens' (see also Hunston & Su, 2019).

We then independently performed the local grammar analyses of all the instances we retrieved and resolved all disagreement. To be more specific, we manually labelled each definitional unit in terms of its communicative functions by using the local grammar terminologies presented in Table 5. The analyses are discussed in more detail below.

For the sake of clarity, we divided our analyses into six sets, according to the degree of prototypicality and similarity. The first set of analyses includes patterns containing those fundamental terminologies for definition, i.e., 'Definiendum', 'Hinge', 'Definiens', and 'Supporting statement'. Some examples are given in Tables 6 and 7.

In these two patterns 'Definiendum' appears in the left part and 'Definiens' in the right part of a definitional clause and the two parts are linked by the element 'Hinge'. The two patterns are realised by, in terms of Systemic Functional Linguistics (Halliday, 1994), relational: identifying clauses. As Halliday (1994, p. 124) argues, "[i]n any identifying clause, the two halves refer to the same thing; but the clause is not a tautology, so there must be some difference between them". The difference, according to Barnbrook and Sinclair (2001, p. 238), is that "one part is more explicit than the other". That is, the syntactic unit labelled 'Definiens' is more specific in meaning than the one labelled 'Definiendum'. The difference between the two patterns lies in that in the second pattern the 'Definiens' is followed by an element labelled 'Supporting statement', providing additional information for the definition.

Patterns in the other five sets of analyses can be regarded as variants of those included in Set 1, in which a definition is made more explicit. For instance, we can specify whether a definition is made by other researcher(s), labelled 'Source-external' (Table 8), or by the author(s) of an article, labelled 'Source-internal' (Table 9), which are categorised respectively into Set 2 and Set 3. At this point, it

**Table 5**
Terminologies for a local grammar analysis of definition.

| Terminology | Explanation | Example |
|---|---|---|
| Definiendum | The term or concept being defined. | ***A proficiency scale*** may be defined as a series … |
| Definiens | The element that indicates how the term or concept is defined. | Lexical bundles have been defined ***as the most frequent recurring lexical sequences*** … |
| Source-internal | The term or concept is being defined by the author(s) him/herself(/themselves). | ***We*** refer to this metric as observed frequency. |
| Source-external | The term or concept is being defined by anyone else other than the author(s). | ***Granger (2002: 11)*** refer to multi-level CLCs as 'quasi-longitudinal' data … |
| Hinge | The element that links different parts of a definition. | Genre ***is*** the context in which … |
| Initiator | The element that indicates that a definition is being given, usually realised by definitional nouns such as *notion, term, concept* etc. | ***The term*** 'interaction' is used to describe the entire online engagement … |
| Scope | The element that indicates the degree to which a definition is valid, usually realised by phrases such as *in this study, in our paper* etc. | ***In this study***, 'EFL context' is defined as learning that occurs … |
| Supporting statement | The element that provides additional information for how a term/concept is being defined, either by giving further specification or by presenting limiting features. | Explicit treatments refer to any instruction ***that contained metalinguistic information*** … |

**Table 6**
Definition realised as Definiendum – Hinge – Definiens.

| Definiendum | Hinge | Definiens |
|---|---|---|
| *Misformation* | *refers to* | *the choice of an inappropriate form of the lemma* |

**Table 7**
Definition realised as Definiendum – Hinge – Definiens – Supporting statement.

| Definiendum | Hinge | Definiens | Supporting statement |
|---|---|---|---|
| *Bootstrapping* | *is* | *an approach to statistical inference* | *that makes fewer assumptions …* |

should be mentioned that, in those instances where there is no clear indication of source, the definition should be regarded as the author's own formulation. In writing practice, this means that EAP learners need to be aware that if they do not specify the source, the definition would count as theirs. In other words, if they draw on notions that have been previously discussed, they must specify the source so as to avoid potential plagiarism.

The fourth set of analyses includes those patterns in which the writers opt to modify a definition by delimiting the extent to which a given definition is valid. The element serving this function is labelled 'Scope' which may occur either in the initial or middle position of a definition sentence, as shown in Tables 10 and 11.

The use of 'Scope' allows the writers to hedge their definition, which echoes the discussion by Pearson (1998) who distinguished two types of hedged definition. One is the 'indicator of tentativeness', which means that the writers are being tentative about their claims, leaving the definition open to challenge or to further refinement at a later stage (Pearson, 1998). The other is 'indicator of scope', which indicates that a definition has only local reference and may not necessarily be applicable on a wider scale (Pearson, 1998). The latter corroborates our label 'Scope' which, in our data, is usually realised by phrases such as *in this paper, in our study, for our purpose*, etc.

Patterns included in Set 5 are those in which the act of definition is marked by definitional nouns (i.e., *concept, definition, notion, term*). In such cases, the definitional noun is labelled 'Initiator', indicating that the act of definition is going to be performed, as shown in Table 12.

The last set of analyses are even more complex, containing two or more additional elements, as shown in Table 13.

In the foregoing text we have discussed the local grammar analyses of definition, which have been divided into six sets. In Table 14, we provide an overview of the 29 local grammar patterns which occur more than once per million words in our corpus, with an example and the quantitative information given for each, whereas the residual seven infrequent patterns are listed in the Appendix.

Overall, our analyses show that local grammars represent a means to make the labelling of syntactic elements more transparent and context-specific, which contributes to the transparency and specificity of the description of discourse acts in general (see also Hunston & Sinclair, 2000; Su & Wei, 2018). It also provides quantitative information for each pattern, indicating the typicality of the ways in which definition can be performed. These would offer important insights into EAP teaching, as will be discussed in the following section.

## 5. The implications and applications of local grammars for EAP writing pedagogy

As noted at the outset of this paper, the present study is part of a larger project which aims to systematically develop local grammars of discourse acts of importance in academic writing and adequately explore the pedagogical value of such local grammars for EAP teaching. The current investigation, zooming in on the discourse act of definition, presents a valuable addition to this line of inquiry. In this section, we further discuss the pedagogical implications and applications of the research for EAP writing pedagogy.

Firstly, research on local grammars of discourse acts could usefully inform EAP course design. McGrath and Kaufhold (2016, p. 4) suggest that in EAP/ESP "course design begins with needs analysis". What learners need, according to Widdowson and Allen (1985, p. 74), is a course which develops "a knowledge of how sentences are used in the performance of different communicative acts" (cited in Pearson, 1998, p. 94). Although Flowerdew (1992, p. 215) has discussed comprehensively "the value of descriptive work on language functions (speech acts) as input to course design" (see also Flowerdew, 2017), little effort has been made to develop courses that use such descriptions as the input for the teaching of discourse or communicative acts. In the case of definition, for example, Darian (2003, p. 47) has pointed out that "[d]efining words and concepts is one of our most important analytical skills, yet a skill that is taught haphazardly, if at all". This calls for a course that offers explicit instruction of discourse acts to facilitate the teaching of academic writing.

We argue that local grammars of discourse acts offer important insights for designing such a course. An easy way to do so is to adapt

**Table 8**
Definition realised as Definiendum – Hinge – Definiens – Source-external.

| Definiendum | Hinge | Definiens | Source-external |
|---|---|---|---|
| *Depth knowledge* | *refers to* | *how well one knows a lexical item* | *(Qian 2002)* |

**Table 9**
Definition realised as Source-internal – Hinge – Definiendum – Definiens – Supporting statement.

| Source-internal | Hinge | Definiendum | Definiens | Supporting statement |
|---|---|---|---|---|
| *Goffman* | *defines* | *an animator* | *as someone* | *who 'animates words he had hand in formulating … '* |

**Table 10**
Definition realised as Scope – Definiendum – Hinge – Definiens.

| Scope | Definiendum | Hinge | Definiens |
|---|---|---|---|
| *In some cases* | *CLIL* | *is defined as* | *a whole program of instruction* |

**Table 11**
Definition realised as Definiendum – Hinge – Scope – Definiens – Supporting statement.

| Definiendum | Hinge | Scope | Definiens | Supporting statement |
|---|---|---|---|---|
| *Chinglish* | *is defined* | *in this study* | *as the expression* | *that is unidiomatic or awkward in English* |

**Table 12**
Definition realised as Initiator – Definiendum – Hinge – Definiens – Supporting statement.

| Initiator | Definiendum | Hinge | Definiens | Supporting statement |
|---|---|---|---|---|
| *A common definition of* | *diagnosis* | *is* | *that it is a formal activity of evaluation and judgement* | *which focuses on problem identification …* |

**Table 13**
Definition realised as Scope – Source-internal – Hinge – Definiendum – Definiens – Supporting statement.

| Scope | Source-internal | Hinge | Definiendum | Definiens | Supporting statement |
|---|---|---|---|---|---|
| *In line with the RS model* | *we* | *define* | *language-based RS* | *as a tendency* | *to anxiously anticipate rejection …* |

existing EAP courses by integrating local grammar descriptions of discourse acts in teaching practice. More ambitiously, attempts can be made to develop an EAP course that specifically applies local grammars to the teaching of discourse acts. Since previous studies have identified a number of discourse acts that are frequently performed in academic writing, such as exemplification, definition, cause and effect, and comparison and contrast (e.g., Jordan, 1999; Kirszner & Mandell, 2017; Paquot, 2008, 2010), the course could be designed to teach systematically these discourse acts, including their lexicogrammatical realisations, discourse-semantic patternings, and the rhetorical effects. Accordingly, each module of the course will be designated for the teaching of one specific discourse act.

This kind of EAP course will need to be equipped with special teaching materials, which leads us to the second pedagogical value underlying our current investigation, i.e., the development of local grammar informed EAP teaching materials. This is based on the observation that, on the one hand, to teach communicative acts we need to make both the linguistic forms and their communicative function as explicit as possible (Pearson, 1998; Widdowson & Allen, 1985). In the case of definition, Dalton-Puffer (2007) rightly argues that explicit instruction and presentation of definition could improve students' ability to define in academic contexts. On the other hand, each local grammar developed of a given discourse act offers an inventory of the lexicogrammatical resources, the associated local grammar patterns, and their respective quantitative information (such as the one of definition shown in Table 14) that are available to perform that specific discourse act. Such an inventory can be fruitfully used as an input to develop materials for EAP teaching, helping teachers to familiarise their students with the realisations of a given discourse act from the most prototypical patterns to the less prototypical yet still appropriate ones. In practice, Su and Zhang (2020) have discussed how findings of local grammars of discourse acts can be applied to develop EAP teaching materials. They suggest that the materials developed for the teaching of a given discourse act may comprise five sections: introducing a given discourse act, retrieving instances realising the target discourse act, performing local grammar analysis, summarising the local grammar patterns, and extra consolidating exercises. Following Su and Zhang (2020) and based on the present study, we argue that the materials for teaching definition can be similarly organised.

Specifically, in Section I we first introduce definition to student and tell them that it is a key discourse act performed to clarify abstract notions or concepts and/or to elaborate technical terms in academic writing. It is also necessary to provide specific scenarios where definitions are used in academic texts to familiarise students with the functions of definitions. In Section II we tell students how to retrieve instances of definition in a corpus. As definition is typically signalled by lexical markers (e.g., *define*, *refer*, *call*) or the construction 'X BE Y' (see Table 4), instances of definition can be retrieved by searching for these markers in a given corpus. In Section III we can invite the students to perform local grammar analyses of definition – the terminologies introduced in the present study could be helpful for this purpose. Based on the analyses, in Section IV we ask the students to summarise the local grammar patterns associated with definition, through which we can establish the local grammar of definition, and then draw the students' attention to the typicality

**Table 14**
An overview of the local grammar of definition in CORAL.

| | Patterns | Raw frequency | Normalised frequency |
|---|---|---|---|
| Set 1 | Definiendum – Hinge – Definiens e.g., *Collocation refers to the co-selection of words.* | 889 | 247.07 |
| | Definiendum – Hinge – Definiens – Supporting statement e.g., *LLAMA is a computer-based test that consists of four subtests designed to measure different aspects of …* | 739 | 205.38 |
| | Definiendum – Definiens – Supporting statement e.g., *Content and Language Integrated Learning (CLIL), an umbrella term for a teaching approach in which content-based subjects are taught …* | 5 | 1.39 |
| | Subtotal | 1633 | 453.84 |
| Set 2 | Source-external – Hinge – Definiendum – Definiens – Supporting statement e.g., *Weick (1995) defines sense-making as the social interactional process in which participants jointly construct meaning.* | 68 | 18.90 |
| | Definiendum – Hinge – Definiens – Source-external e.g., *Extrinsic modality refers to assessments of likelihood (Biber* et al. *1999: 485).* | 47 | 13.06 |
| | Source-external – Hinge – Definiendum – Definiens e.g., *They defined fluency as "the rapid production of language".* | 33 | 9.17 |
| | Source-external – Definiendum – Hinge – Definiens – Supporting statement e.g., *According to Carroll and Sapon (2002), language aptitude refers to a set of cognitive abilities that are 'predictive of how well … '* | 32 | 8.89 |
| | Definiendum – Hinge – Source-external – Definiens – Supporting statement e.g., *An event-frame is defined by Talmy (2000: 259) as 'a set of conceptual elements and interrelationships that are evoked together or co-evoke each …* | 13 | 3.61 |
| | Definiendum – Hinge – Definiens – Supporting statement – Source-external e.g., *'episodic evidence', that is, the activities a person happens to be engaged in (Clark 1996b).* | 7 | 1.95 |
| | Source-external – Hinge – Definiendum – Hinge – Definiens – Supporting statement e.g., *Miller and Shepherd (2002) use it also refer to the time at which a new genre emerges.* | 6 | 1.67 |
| | Definiendum – Hinge – Source-external – Hinge – Definiens – Supporting statement e.g., *… the term input enhancement which he defined as a process whereby input becomes salient to the learner.* | 5 | 1.39 |
| | Source-external – Definiendum – Hinge – Definiens e.g., *In Levelt's (1989) model, speech production consists of three stages of interactive and overlapping processing.* | 5 | 1.39 |
| | Source-external – Hinge – Definiendum – Hinge – Definiens e.g., *Van Patten (2009) has used the term 'processing instruction' to refer to structured input activities …* | 4 | 1.11 |
| | Subtotal | 220 | 61.14 |
| Set 3 | Source-internal – Hinge – Definiendum – Definiens – Supporting statement e.g., *This text defines 'bilingual education' as a method which develops and maintains first language literacy …* | 23 | 6.39 |
| | Source-internal – Hinge – Definiens – Definiendum e.g., *We refer to this metric as dispersion.* | 18 | 5.00 |
| | Definiens – Hinge – Source-internal – Hinge – Definiendum e.g., *… these two metonymic shifts are what we call regular metonymy.* | 15 | 4.17 |
| | Definiendum – Source-internal – Hinge – Definiens e.g., *… the hypostatization of 'ELF' initiates at the level of ideology what I refer to as 'lingua franca fetishism'.* | 10 | 2.78 |
| | Definiendum – Source-internal – Hinge – Definiens – Supporting statement e.g., *By value I mean the prestige and importance that is ascribed to English …* | 7 | 1.95 |
| | Source-internal – Hinge – Definiendum – Hinge – Definiens – Supporting statement e.g., *I use the label 'constructionist' to refer to the central notion that interviews are co-constructed.* | 6 | 1.67 |
| | Subtotal | 79 | 21.96 |
| Set 4 | Scope – Definiendum – Hinge – Definiens e.g., *In this paper, 'language ideology' refers both to a concept and a field of study.* | 59 | 16.40 |
| | Scope – Definiendum – Hinge – Definiens – Supporting statement e.g., *In this article, age of acquisition (AoA) refers to the age at which immersion in the L2 begins.* | 46 | 12.78 |
| | Definiendum – Hinge – Scope – Definiens – Supporting statement e.g., *Chinglish is defined in this study as the expression that is unidiomatic or awkward in English …* | 10 | 2.78 |
| | Definiendum – Scope – Hinge – Definiens – Supporting statement e.g., *Reactivity in this context refers to the effect, be it positive or negative, that verbalizing one's thoughts has on the processes …* | 9 | 2.50 |
| | Definiendum – Scope – Hinge – Definiens e.g., *The 'management' in corporate language management refers to the management of language-related issue.* | 8 | 2.22 |
| | Subtotal | 132 | 36.68 |
| Set 5 | Initiator – Definiendum – Hinge – Definiens – Supporting statement e.g., *The concept 'frame' is often used for describing the way our knowledge is organised and how we use our knowledge …* | 19 | 5.28 |
| | Initiator – Definiendum – Hinge – Definiens e.g., *The term spillover effect is used to describe the sustained processing of a target word.* | 11 | 3.06 |
| | Subtotal | 30 | 8.34 |
| Set 6 | Scope – Source-internal – Hinge – Definiens – Definiendum e.g., *In the present article, we will refer to such recurrent word combinations as multi-word expressions (MWEs).* | 16 | 4.45 |
| | Scope – Source-internal – Hinge – Definiendum – Definiens – Supporting statement e.g., *In line with the RS model, we define language-based RS (language-RS) as a tendency to anxiously anticipate rejection by the dominant …* | 8 | 2.22 |
| | Initiator – Definiendum – Hinge – Source-external – Definiens – Supporting statement e.g., *A prominent definition of stance is given by Du Bois as 'a public act by a social actor, achieved dialogically through overt communicative means … '.* | 4 | 1.11 |
| | Subtotal | 28 | 7.78 |
| | **Total** | **2122** | **589.75** |

of the local grammar patterns as well as the rhetorical functions of the constituting elements. Finally, in Section V we can provide learners with extra consolidating exercises and design prompts on essential concepts relevant to their disciplinary subjects so as to elicit their performance in definitions. In this way, the students are encouraged to embed the learnt local grammar patterns of definition into their own writing, which could in turn enable them to define in ways that are acceptable to the community of their chosen discipline.

Above we discussed how to develop local grammar informed EAP teaching materials for the discourse act of definition. Arguably, however, pedagogical materials for teaching all kinds of discourse acts can be organised in such ways. While little effort has been made to systematically develop this kind of EAP teaching materials, Zhang and Su (2021) have made a trial of teaching discourse acts to advanced EFL learners by using the local grammar based EAP teaching materials and found that the local grammar approach could indeed facilitate EAP teaching and learning. This suggests that more efforts need to be made to develop local grammar informed EAP teaching materials and apply such materials in EAP writing pedagogy.

What we have proposed here are general guidelines for EAP teaching practitioners to flexibly incorporate local grammars of discourse acts into EAP course design and material development. Despite our narrow focus on the discipline of Linguistics in this paper, we believe that the implications drawn here can be extended to a wider range of disciplinary contexts, as academic writing in all disciplines involves the use of such discourse acts or functions (Dalton-Puffer, 2013, 2016; Jordan, 1999; Nashaat-Sobhy & Llinares, 2020; Paquot, 2010). It should be cautioned, though, that the implementation of the proposed local grammar informed EAP pedagogy will need to accommodate different disciplinary needs, as "[n]umerous studies now show the extent to which language features are specific to particular disciplines and that the best way to prepare students for their studies is […] to provide them with an understanding of the features of the discourses they will encounter in their particular courses" (Hyland, 2008, p. 20). This means that, in classroom teaching, the EAP instructors need to use a corpus that is representative of a particular discipline, and then guide the students to engage in subsequent local grammar analysis, which could ultimately help develop students' ability to meet the linguistic demands of discipline-specific writing.

Finally, it merits mentioning that the local grammar approach to teaching discourse acts is highly compatible with the discussion of Cognitive Discourse Functions (CDFs) in the context of CLIL (content and language integrated learning; Dalton-Puffer, 2013, 2016; Morton, 2020) in that both emphasise the role that discourse functions play in the development of student or L2 writers' academic literacy. Although CDFs are a very useful framework for tidying up the learning objectives that allow students to engage with disciplinary contents, "they do not stipulate *exactly* which language (at the levels of lexis and grammar) students have to use" (Morton, 2020, p. 11; italics in original). In this respect, each local grammar, by offering an inventory of the lexicogrammatical resources that are available to perform a given discourse function, could be usefully incorporated into CLIL to enrich both the teachers' and students' repertories of linguistic choices and enable them to teach and perform rhetorical functions in discipline-specific ways (cf. Walsh Marr et al., 2021).

## 6. Conclusion

The present study reported an attempt to develop a local grammar of definition in Linguistics research articles, aiming to contribute to the overarching goal of systematically developing local grammars of discourse acts of importance in academic writing. It has been shown that each local grammar provides a transparent and specialised description of the target discourse act, which in turn offers an inventory of linguistic resources for performing that discourse act. The pedagogical implications and applications of the resulting local grammars of discourse acts have also been discussed in detail, especially in EAP course design and the development of local grammar informed EAP teaching materials. Overall, the study has shown that local grammars can be a useful approach to explore discourse acts of importance in academic writing and findings of such research offers fresh insights into EAP instruction, thereby contributing significantly to EAP writing research and pedagogy.

The study not only points to the need to develop local grammars of other discourse acts in academic context, but also opens up new avenues for future investigation into the specific discourse act of definition. For instance, future research can employ the local grammar approach to investigate the extent to which novice or L2 writers' use of definitions is similar to or different from that of expert writers (cf. Su et al., 2022). Moreover, given that language use has been shown to vary according to disciplines (Charles, 2006; Hyland, 2008; Jiang, 2017; Lu et al., 2021), it would be worthwhile to explore how definitions are performed by writers in different disciplines (cf. Triki, 2019). Additionally, it would also be worthwhile to explore how definitions (and by extension other discourse acts) are distributed across different sections in research articles. Investigation into these issues could enhance our understanding of how definition works in academic writing and can be taught in the EAP context.

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## CRediT authorship contribution statement

**Yimin Zhang:** Conceptualization, Data curation, Data collection, Formal analysis, Data analysis, Writing – original draft, Writing – review & editing. **Hang Su:** Conceptualization, Data curation, Data collection, Formal analysis, Data analysis, Funding acquisition, Project administration, Writing – review & editing.

## Declaration of competing interest

None.

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## Appendix. Patterns occurring less than 1 time per million words

| Patterns | Raw frequency | Normalised frequency |
|---|---|---|
| Initiator – Definiendum – Definiens – Supporting statement – Source-external e.g., … *the concept of genre awareness/ knowledge: writers' ability to orchestrate and flexibly deploy rhetorical resources across contexts and languages (Hyland, 2007; Johns 2008; Tardy 2009)*. | 3 | 0.83 |
| Initiator – Source-external – Definiendum – Hinge – Definiens – Supporting statement e.g., *Following the definition provided by Gardner* et al. *(1997) of motivational intensity as the goal-directed effort that learners expend to learn a foreign language and their persistence in learning …* | 2 | 0.56 |
| Initiator – Definiendum – Hinge – Definiens – Supporting statement – Source-external e.g., *The term 'ethnic media' is used broadly in the literature to include media by and for (i) immigrants,(ii) ethnic, racial, and linguistic minorities, and (iii) indigenous populations across different parts of the world (Matsaganis et al. 2011: xiii)*. | 1 | 0.28 |
| Scope – Source-external – Hinge – Definiendum – Definiens – Supporting statement e.g., *Coming from a literacy and writing studies context, Kress defines mode as a socially and culturally shaped resource for meaning-making.* | 1 | 0.28 |
| Source-internal – Definiendum – Hinge – Definiens e.g., *Thus, our definition of collocational accuracy refers to how accurate learners are in producing acceptable and expected multiword units.* | 1 | 0.28 |
| Scope – Source-internal – Hinge – Definiendum – Hinge – Definiens e.g., *Throughout the article, I use the term 'speaker' to refer to both speakers and writers.* | 1 | 0.28 |
| Source-internal – Hinge – Definiendum – Scope – Definiens e.g., *I use the term 'collocation' generally in this article to include both 'colligation' and 'collocation'*. | 1 | 0.28 |

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